

PS200C: Quantitative Methods

In-class Activity: Selection on Observables

Name(s): _____

Question 1: The identification argument

Suppose you assume conditional ignorability: $\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i \mid X_i$.

(a) In one sentence, what does this assumption say in plain English?

(b) What are some other names for this assumption?

Question 2: Evaluating SOO claims

For each study, state what the SOO/CI assumption would require, and whether you find it plausible.

(a) A researcher estimates the effect of college attendance on earnings by comparing college-goers to non-goers, “controlling for” high school GPA, parental income, and state of residence.

(b) Blattman & Annan (2009) estimate the effect of abduction by the LRA on education, conditioning on village location and age, arguing that abduction was quasi-random conditional on these variables.

Question 3: Post-treatment variables

A randomized job training program (D) aims to increase earnings (Y). A researcher notices that some treated individuals dropped out of the program early, and restricts the analysis to “completers” vs. controls.

- (a) Is “program completion” a pre-treatment or post-treatment variable?
- (b) Why could restricting to completers bias the estimate, even though D was randomly assigned?
(Think about what kinds of people complete vs. drop out.)

Question 4: Subclassification

You have two strata defined by a covariate $X \in \{\text{low}, \text{high}\}$:

Stratum	$\bar{Y}$ (treated)	$\bar{Y}$ (control)	N_j (total)
$X = \text{low}$	12	8	60
$X = \text{high}$	20	18	40
Total			100

- (a) Compute the subclassification estimate of the ATE.
- (b) Suppose the treated group is disproportionately in the “high” stratum. Would you expect the unadjusted DIM to overestimate or underestimate the ATE? Why?
- (c) Under what assumption is your answer to (a) a valid causal effect?